Chapter 4

H Chan

Exercise 1.

Let k be an algebraically closed field of characteristic not 2, $f \in k[x_1]$ a nonconstant polynomial, $Y = V(x_2^2 - f) \subset \mathbb{A}_k^2$ and $\phi : Y \to \mathbb{A}_k^1$ the morphism given by $(x_1, x_2) \mapsto x_1$. Show that ϕ is a finite morphism that is étale over the point of $\mathbb{A}_k^1$ corresponding to $a \in k$ if and only if $f(a) \neq 0$.

Proof.

ϕ corresponds to the ring homomorphism $k[x] \to k[x, y]/(y^2 - f(x))$ where $x \mapsto x$. ϕ is indeed a finite morphism as $k[x, y]/(y^2 - f(x))$ generated by the set $\{1, y\}$ as a $k[x]$-module. Then ϕ is étale over a point a if and only if $k[x, y]/(y^2 - f(x), x - a)$ is an étale $k[x]/(x - a)$-algebra where $k[x]/(x - a) \cong k$. But then $k[x, y]/(y^2 - f(x), x - a) \cong k[y]/(y^2 - f(a))$ is étale over k if and only if $f(a) \neq 0$; as $y^2 - f(a)$ always splits with distinct factors if $f(a) \neq 0$ because k is algebraically closed with characteristic not 2. ■

Exercise 2.

Let k be an algebraically closed field of characteristic 0, and $\phi : Y \to \mathbb{A}^1_k$ a finite surjective morphism over k such that $\mathcal{O}(Y) \cong \mathcal{O}(\mathbb{A}^1_k)[f]$ with some $f \in \mathcal{O}(Y)$. Verify directly that ϕ cannot be étale above the whole of $\mathbb{A}^1_k$.

Proof.

The question statement is only true if one further assumes that ϕ has degree at least 2; i.e. not an isomorphism.

Proof 1.

As ϕ is a finite morphism, $\mathcal{O}(Y)$ must be a finitely generated $\mathcal{O}(X)$-module such that there exists some irreducible $p(x,t) \in \mathcal{O}(\mathbb{A}^1_k)[t] \cong k[x][t]$ with $\mathcal{O}(Y) \cong k[x][f] \cong k[x,t]/(p(x,t))$. Hence

$$\begin{aligned}\phi \text{ is étale} &\iff k[t]/(p(a,t)) \text{ is an étale } k\text{-algebra for all } a \\ &\iff p(a,t) \text{ is separable for all } a \\ &\iff \Delta(p(a,t)) \neq 0 \text{ for all } a.\end{aligned}$$

But then this is a first order statement, such that by the Lefschetz principle it is sufficient to prove the case $k = \mathbb{C}$.

If ϕ were to be étale above all of $\mathbb{A}^1_{\mathbb{C}}$, by Proposition 4.5.6 Y induces a finite unbranched cover of $\mathbb{C}$. But then $\mathbb{C}$ is simply connected such that Y must be a trivial cover. As irreducible curves are connected this implies that Y is necessarily trivial, so ϕ is an isomorphism. ■

Proof 2.

By Proposition 4.4.6, the morphism $Y \to \mathbb{A}^1_k$ can be extended into a morphism of projective curves $\phi' : \overline{Y} \to \mathbb{P}^1_k$. As $\operatorname{char}(k) = 0$ the ramification of ϕ' is automatically tame. Thus if ϕ is étale everywhere, ϕ' must be ramified above ∞ only. We show a contradiction similar to as done in Chapter 3 Exercise 4: By Hurwitz's formula (Hartshrone Ch. IV.2 Corollary 2.4, p.301) we have that

$$2g\left(\overline{Y}\right) - 2 = n \cdot (2g(\mathbb{P}^1_k) - 2) + \sum_{P \in Y} (e_P - 1) \leq -2n + n - 1$$

such that $0 \leq g\left(\overline{Y}\right) \leq \frac{1-n}{2}$. But then this is not possible for $n \geq 2$. ■

Exercise 3.

Let k be an algebraically closed field, and n an integer prime to the characteristic of k. Prove in a purely algebraic way that up to isomorphism there is a unique cyclic Galois cover of $\mathbb{A}^1_k$ with group $\mathbb{Z}/n\mathbb{Z}$ étale outside 0, as in Example 4.6.12 (2). [*Hint:* Use Kummer theory (Remark 1.2.10).]

Proof.

By Remark 1.2.10, every $\mathbb{Z}/n\mathbb{Z}$-Galois extension of $\mathcal{K}(\mathbb{A}^1_k) = k(x)$ is generated by some nth root as k admits all nth roots of unity.

Hence let $k(x)\left[\sqrt[n]{f}\right]$ denote such an extension, and as k is algebraically closed we can WLOG write $f = \prod (x-\alpha_i)^{n_i}$. Then the associated ring of regular functions $\mathcal{O}(Y)$ is exactly the integral closure of $\mathcal{O}(\mathbb{A}^1_k) = k[x]$, which is seen to clearly contain $k[x]\left[\sqrt[n]{f}\right] \cong k[x,t]/(t^n - f)$.

In particular, the ring $\mathcal{O}(Y)/(x-\alpha_i)\mathcal{O}(Y)$ contains the k-algebra $k[x,t]/(x-\alpha_i, t^n - f) \cong k[t]/(t^n)$ which is not étale. As every subalgebra of an étale algebra is necessarily étale, $\mathcal{O}(Y)$ cannot be étale above α_i. Thus $f = x^k$ exactly. But then as f has to be an nth root, k has to be coprime to n such that $\mathcal{K}(Y) = k(x)\left[\sqrt[n]{x^k}\right] = k(x)\left[\sqrt[n]{x}\right]$ exactly. Hence by Theorem 4.3.10 we are guaranteed uniqueness of the cover.

It remains to verify that the cover Y corresponding to $\mathcal{K}(Y) = k(\sqrt[n]{x})$ is indeed étale outside 0. To that end, notice that the ring $k[x]\left[\sqrt[n]{f}\right] = k\left[\sqrt[n]{x}\right]$ is now a UFD as it is isomorphic to $k[t]$. Hence it is integrally closed, and $\mathcal{O}(Y) = k\left[\sqrt[n]{x}\right]$. It can now be easily verified as in Example 4.5.4 that this is étale outside of 0. ■

Exercise 4.

Let k be an algebraically closed field of characteristic $p > 0$, and consider the rational function field $k(t)$. Choose $f \in k[t]$ so that the polynomial $x^p - x - f \in k(t)[x]$ is irreducible over $k(t)$.

(a) Show that the normalization of $\mathbb{A}^1_k$ in the finite extension of $k(t)$ defined by $x^p - x - f$ is an étale Galois cover of $\mathbb{A}^1_k$ whose Galois group is $\mathbb{Z}/p\mathbb{Z}$; it is called an *Artin–Schreier cover* of $\mathbb{A}^1_k$.

Proof.

That the extension is $\mathbb{Z}/p\mathbb{Z}$-Galois is a route verification (the isomorphism is given by $k \mapsto \sigma_k(\overline{x}) = \overline{x} + k$). By Theorem 4.3.10, this corresponds to a finite separable morphism $Y \to \mathbb{A}^1_k$, where $\mathcal{O}(Y)$ is the normalization of $\mathbb{A}^1_k$ in the extension. It remains to show that the morphism is étale over a nonempty open set U.

Write $f = \prod (t - \alpha_i)^{n_i}$ (where $i > 0$ as k is algebraically closed), and take $U = \mathbb{A}^1_k$. By the same reasoning as in Exercise 3, $\mathcal{O}(Y)/(t-a)\mathcal{O}(Y)$ contains the k-algebra $k[t,\overline{x}]/(t-a) \cong k[x]/(x^p - x - f(a))$. For $a \in U$, the polynomial $q(t) = x^p - x - f(a)$ has p distinct roots (it is separable as $\gcd(q, q') = \gcd(q, -1) = 1$). Hence $k[x]/(x^p - x - f(a)) \cong k^p$ is a dimension p k-vector space, and in particular is étale. But then by the ramification-inertia identity (Proposition 4.1.6) this must be the whole of $\mathcal{O}(Y)/(t-a)\mathcal{O}(Y)$, as it must have dimension equal to the degree of the field extension. Thus Y is étale above U. ■

(b) Conclude that the maximal abelian quotient of exponent p of $\pi_1(\mathbb{A}^1_k)$ is an infinite dimensional $\mathbb{F}_p$-vector space of dimension equal to the cardinality of k.

Proof.

It is sufficient to determine the dimension of the group as an $\mathbb{F}_p$-vector space. To that end we count the number of possible $\mathbb{Z}/p\mathbb{Z}$ extensions of $k(t)$, which we know to be generated by a root of a polynomial of the form $x^p - x - f$ for some $f \in k(t)$ by Remark 1.2.10.

By (a), $f := t - a$ for $a \in k$ gives at least $|k|$ many such polynomials. Indeed if $\alpha \in k(t)$ is a root of $x^p - x - f$ then $\mathrm{val}_t(x^p - x) = p \cdot \mathrm{val}_t(x)$ but $\mathrm{val}_t(f) = 1$.

On the other hand, there are at most $|k(t)|$ many possible choices of f that would generate a $\mathbb{Z}/p\mathbb{Z}$ extension. But then $|k(t)| \leq |k[t] \times k[t]|$, and k being algebraically closed implies that $|k|$ is at least countably infinite; such that $|k(t)| = |k[t] \times k[t]| = |k[t]| = |k|$. ■

Exercise 5.

Give an example showing that the converse statement in Lemma 4.5.10 fails if one does not assume that Y is normal.

[*Hint:* Take $X = \mathbb{A}^1_k$, $Y = V(x_2^2 - x_1^3 + x_1^2) \subset \mathbb{A}^2_k$, Z the normalization of Y and $\phi : (x_1, x_2) \mapsto x_1$.]

Proof.

For ease of notation we replace x_1 by x and x_2 by y. For simplicity assume the characteristic is 0 (it is, however, sufficient for it to not be 2 in the following proof).

We first compute the integral closure of $\mathcal{O}(Y)$: Notice that $y/x \in \mathcal{K}(Y)$ is integral over $\mathcal{O}(Y) = k[x,y]/(y^2 - x^3 + x^2)$; as $(y/x)^2 - x + 1 = 0$. So $\mathcal{O}(Z)$ contains the algebra $k[x,y]/(y^2 - x + 1)$. But then $k[x,y]/(y^2 - x + 1) \cong k[x,y]/(y^2 - x) \cong k[\sqrt{x}]$ is a UFD, so we get equality.

Thus the composite map $Z \to X$ is a morphism of integral affine normal curves, given by the ring homomorphism $k[x] \hookrightarrow k[x,y]/(y^2 - x + 1)$. This morphism can be seen to be étale everywhere except $x = 1$ (if $\mathrm{char}(k) = 2$ then this morphism is ramified everywhere).

On the other hand, we see that the morphism $Y \to X$ is not étale at the point $x = 0$: The corresponding k-algebra is $k[x,y]/(x, y^2 - x^3 + x^2) = k[y]/(y^2)$ which is not étale. ■

Exercise 6.

Let k be an algebraically closed field, X an integral proper normal curve over k, and $f \in K(X)$ a rational function.

(a) Verify that the morphism $\phi_f : X \to \mathbb{P}^1_k$ defined in Section 4.4 satisfies $\phi_f(P) = f(P)$ for all closed points $P \in X$ where $f \in \mathcal{O}_{X,P}$, and $\phi_f(P) = \infty$ otherwise.

Proof.

ϕ_f is defined to be the morphism induced from the field extension $k(f) \subseteq K(X)$ as in the construction of Proposition 4.4.5. In particular, it is the morphism that sends the DVR $\mathcal{O}_{X,P} \subseteq K(X)$, associated with a point $P \in X$, to the DVR $\mathcal{O}_{X,P} \cap k(f)$, associated with $\phi_f(P) \in \mathbb{P}^1_k$.

Suppose that $P \in X$ such that $f \in \mathcal{O}_{X,P}$. Then $\mathcal{O}_{X,P} \cap k(f)$ clearly contains $k[f]$, and as $f - f(P)$ vanishes at P we have that $f - f(P) \in \mathfrak{m}_P$ such that $(f - f(P))^{-1} \in \mathcal{O}_{X,P}$. Thus $\mathcal{O}_{X,P} \cap k(f)$ contains the local ring $k[f]_{f-f(P)}$, which is seen to be the DVR corresponding to the point $f(P) \in \mathbb{P}^1_k$.

Now suppose that $P \in X$ such that $f \notin \mathcal{O}_{X,P}$. Then $f^{-1} \in \mathcal{O}_{X,P}$ necessarily such that $f^{-1}(P) = 0$. Then the same argument as above gives us that the corresponding DVR is $k[f^{-1}]_{f^{-1}}$, corresponding to the point ∞. ■

(b) Show that ϕ_f is étale exactly above those points whose preimages satisfy $(f'/f)(P) \neq 0$.

Proof.

The question as stated is incorrect - the z^2-double cover of $\mathbb{P}^1_{\mathbb{C}}$ is ramified at 0 but $f'/f = 2$. The correct statement is that $f'(P) \neq 0$ *or* $(1/f)'(P) \neq 0$.

First assume that $f(P) \in k$. Then by Lemma 4.5.2, f is étale above P if only if $f - f(P)$ lifts to a uniformizer for all the points in the fiber of P (separability is automatic as the residue fields are necessarily both k itself). But then that is true if and only if $f'(P) \neq 0$, as $f - f(P)$ would otherwise land in $\mathfrak{m}_P^2$.

For $P = \infty$, by post-composing with the automorphism $z \mapsto z^{-1}$ we can replace f with $1/f$ such that $1/f(P) = 0 \in k$. The exact same reasoning as above tells us that ϕ_f is étale above ∞ if and only if $(1/f)'(P) \neq 0$. ■

Exercise 7.

Prove the following sharpening of Belyi's theorem: an integral proper normal curve over an algebraically closed field k of characteristic 0 can be defined over $\overline{\mathbb{Q}}$ if and only if there is a morphism $p : X \to \mathbb{P}^1_k$ étale over $\mathbb{P}^1_k \setminus \{0, 1, \infty\}$ such that all ramification indices at points lying above 1 are equal to 2. [*Hint:* Compose with an appropriate Belyi function.]

Proof.

By the statement of Belyi's theorem as given (Theorem 4.7.6), it is sufficient to show that every morphism $p : X \to \mathbb{P}^1_k$ étale over $\mathbb{P}^1_k \setminus \{0, 1, \infty\}$ can be refined to a morphism that also has ramification index 2 above $z = 1$.

To that end, we choose $A = B = 1$, corresponding to the Belyi function $\phi = x(x-1)$. By the same argument as given in Theorem 4.7.6, ϕ is étale outside $\phi(\{0, 1, \infty\}) = \{0, \infty\}$ along with the images of the points where the derivative ϕ' vanishes (namely $\phi(1/2) = -1/4$).

We compute the ramification of ϕ above $-1/4$: The corresponding ring extension is $\mathbb{Q}[x(x-1)] \to \mathbb{Q}[x][(x(x-1))^{-1}]$. The point $1/4$ corresponds to the maximal ideal $\langle x(x-1) + 1/4\rangle = \langle x^2 - x + 1/4\rangle$, which decomposes into $\langle x - 1/2\rangle^2$ in the ring extension. So ϕ has ramification index 2 above the point $-1/4$.

We can conclude by composing with the affine transformation $z \mapsto -4z$ to send $-1/4$ to 1, and we are done. ■

Exercise 8.

Prove that when G is a finite group with trivial center having a rigid system $(C_1, ..., C_n)$ of not necessarily rational conjugacy classes, the conclusion of Theorem 4.8.7 holds with $\mathbb{Q}$ replaced by $\mathbb{Q}(\mu)$, where μ is the group of all complex roots of unity.

Proof.

Exercise 9.

Verify that for $n \geq 3$ the three conjugacy classes in the symmetric group S_n consisting of 2-cycles, $(n-1)$-cycles and n-cycles, respectively, form a rigid system of rational conjugacy classes. [*Remark:* It then follows from Theorem 4.8.7 that S_n occurs as a Galois group over $\mathbb{Q}(t)$. For a more elementary proof, see [93], Section 4.4.]

Proof.

We first verify that these conjugacy classes are rational. To that end, it is sufficient to notice that an l-th power of a k-cycle is still a k-cycle if and only if $\gcd(k,l) = 1$. But then $2, n-1$ and n are all factors of $|S_n| = n!$, so they indeed form rational conjugacy classes.

We next give an explicit instance of a generating 3-tuple, namely $(1\ \ n)(2\cdots n)^{-1}(1\cdots n) = e$. The rest of this proof is dedicated to proving group-theoretic lemmas that will show that the chosen conjugacy classes are rigid.

We prove that $\{(1\ \ n), (1\cdots n)\}$ is a generating set for S_n: By relabelling WLOG assume we are working with the set $\{(1\ \ 2), (1\cdots n)\}$ instead. We will show that $\langle (1\ \ 2), (1\cdots n)\rangle$ contains every transposition. To that end, notice that $(ab) = (1b)(1a)(1b)$ so it is sufficient to show that $\langle (1\ \ 2), (1\cdots n)\rangle$ contains every transposition of the form $(1k)$. But then one can verify that $(1k) = [(12)(12...n)]^{-(n-k+1)} \circ (12) \circ [(12)(12...n)]^{n-k+1}$.

We verify that S_n acts transitively on the set of all generating 3-tuples (with the chosen conjugacy classes). But then as conjugation in S_n are in correspondence with relabelling of the numbers, the same argument as above applies to show that every 3-tuple in the orbit of $\{(1\ \ n), (2\cdots n)^{-1}, (1\cdots n)\}$ is a generating set. ■

Exercise 10.

Let G be a finite group with trivial center possessing a rigid system of three rational conjugacy classes, and let $H \subset G$ be a subgroup of index 2. (Example: $G = S_n$, $H = A_n$, $n \geq 3$.) Show there exists a Galois extension of $\mathbb{Q}(t)$ regular over $\mathbb{Q}$ with group H.

[*Hint:* Apply Theorem 4.8.7 to get a Galois extension $K|\mathbb{Q}(t)$ with group G corresponding to a morphism étale outside three $\mathbb{Q}$-rational points, and set $L := K^H$. Use the Riemann–Hurwitz formula (Corollary 3.6.12) to show that the morphism $X \to \mathbb{P}^1_{\mathbb{Q}}$ corresponding to $L|\mathbb{Q}(t)$ is étale outside exactly two points, and conclude that $L \cong \mathbb{Q}(u)$ for some $u \in K(X)$.]

Proof.

Exercise 11.

(Tamagawa, Koenigsmann) Let X be a proper normal curve over a perfect field k. Fix a separable closure K_s of the function field K of X, let $K_X \subset K_s$ be the Galois subextension with Galois group $\pi_1(X)$ over K, and $\overline{k}$ the algebraic closure of k in K_X.

(a) Let s be a section of the natural projection $\pi_1(X) \to \mathrm{Gal}\big(\overline{k} \mid k\big)$, and denote by D the image of s in $\pi_1(X)$; it has a natural action on K_X. Prove the equivalence of the following statements about a k-rational point P of X:

- D is the decomposition group of a closed point of the pro-étale cover $\tilde{X} \to X$ of Remark 4.6.6 lying above P.
- For all finite extensions $L \mid K$ contained in the fixed field of D the corresponding proper normal curve has a k-rational point lying above P.

(b) Show that the surjectivity part of Grothendieck's Section Conjecture (Conjecture 4.9.9) is equivalent to the following statement: if k is finitely generated over $\mathbb{Q}$, then for *every* proper normal curve X of genus at least 2 the projection $\pi_1(X) \to \mathrm{Gal}\big(\overline{k} \mid k\big)$ has a section if and only if X has a k-rational point. [*Hint:* Use the generalization of the theorem of Faltings ([49], Chapter I, Corollary 2.2) according to which such an X has finitely many k-rational points.]